Time Series Forecasting — Engineering Report

Project 009 · ARIMA Forecasting for Datacenter Power & Silicon Aging

Project Overview

This project implements **ARIMA time series forecasting** across two domains: datacenter power demand prediction (17,520 hourly readings) and silicon transistor aging prediction (1,095 daily Vth shift measurements). The pipeline includes seasonal decomposition, stationarity testing (ADF), ACF/PACF analysis for parameter selection, and multi-step forecasting.

Datasets

Dataset	Rows	Frequency	Target	Domain
Datacenter Power Demand	17,520	Hourly	power_kw	Cloud Infrastructure
Silicon Aging Vth Shift	1,095	Daily	vth_shift_mv	Semiconductor Reliability

Methodology

- **Decomposition:** Trend + Seasonal + Residual extraction
- **Stationarity:** ADF test, differencing (d=1)
- **Parameter Selection:** ACF/PACF analysis → ARIMA(5,1,0)
- **Model:** statsmodels ARIMA with maximum likelihood estimation
- **Evaluation:** AIC (Akaike Information Criterion)
- **Serialization:** joblib export (arima_datacenter.pkl, arima_silicon_aging.pkl)

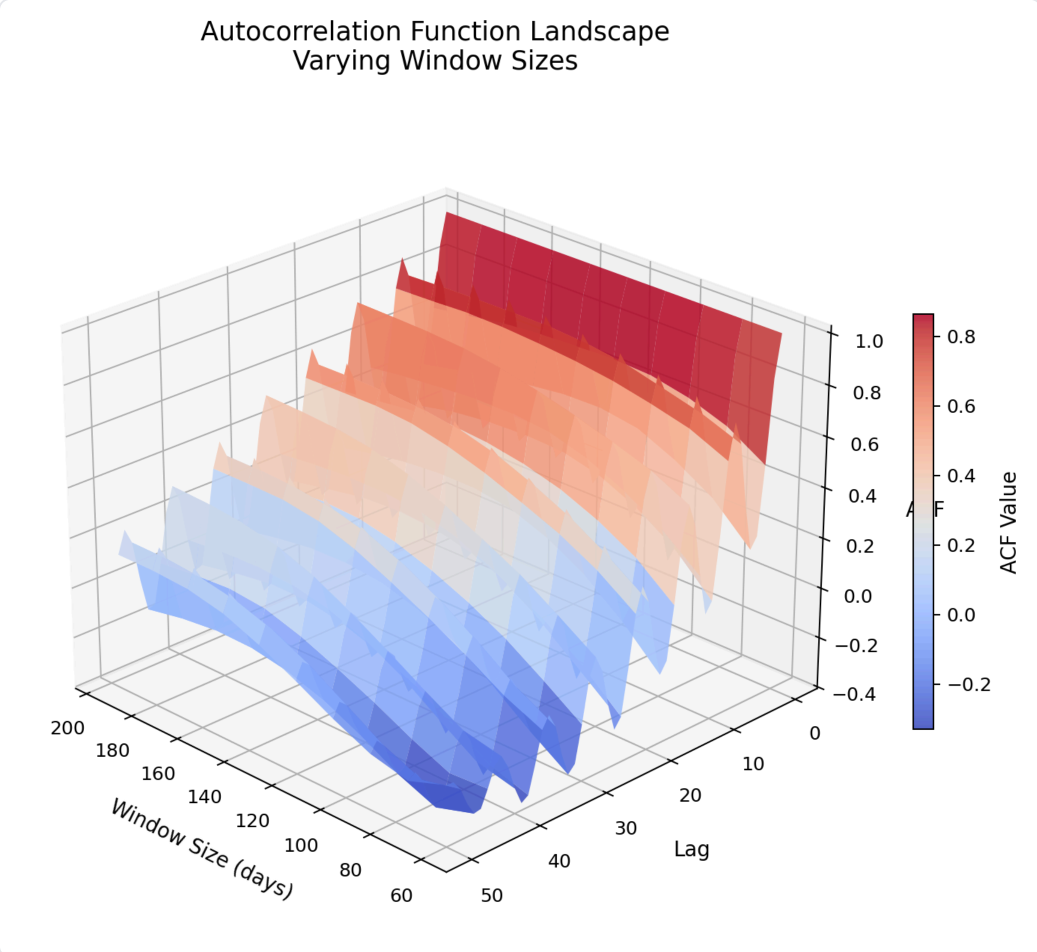
Results

Datacenter AIC: 184,514

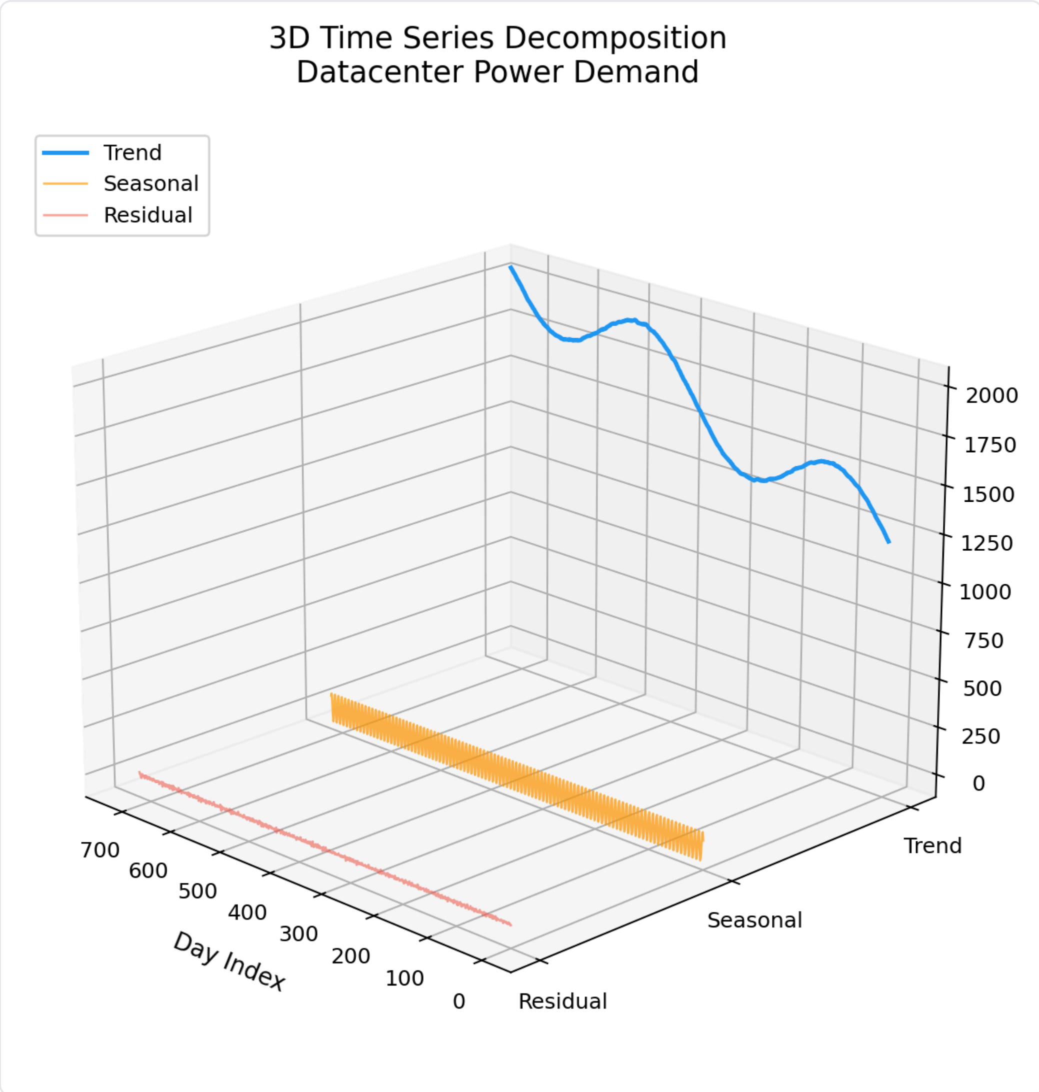
Silicon Aging AIC: 1,772

Lower AIC indicates better model fit relative to complexity. The silicon aging model achieves much lower AIC due to smoother, more predictable degradation patterns.

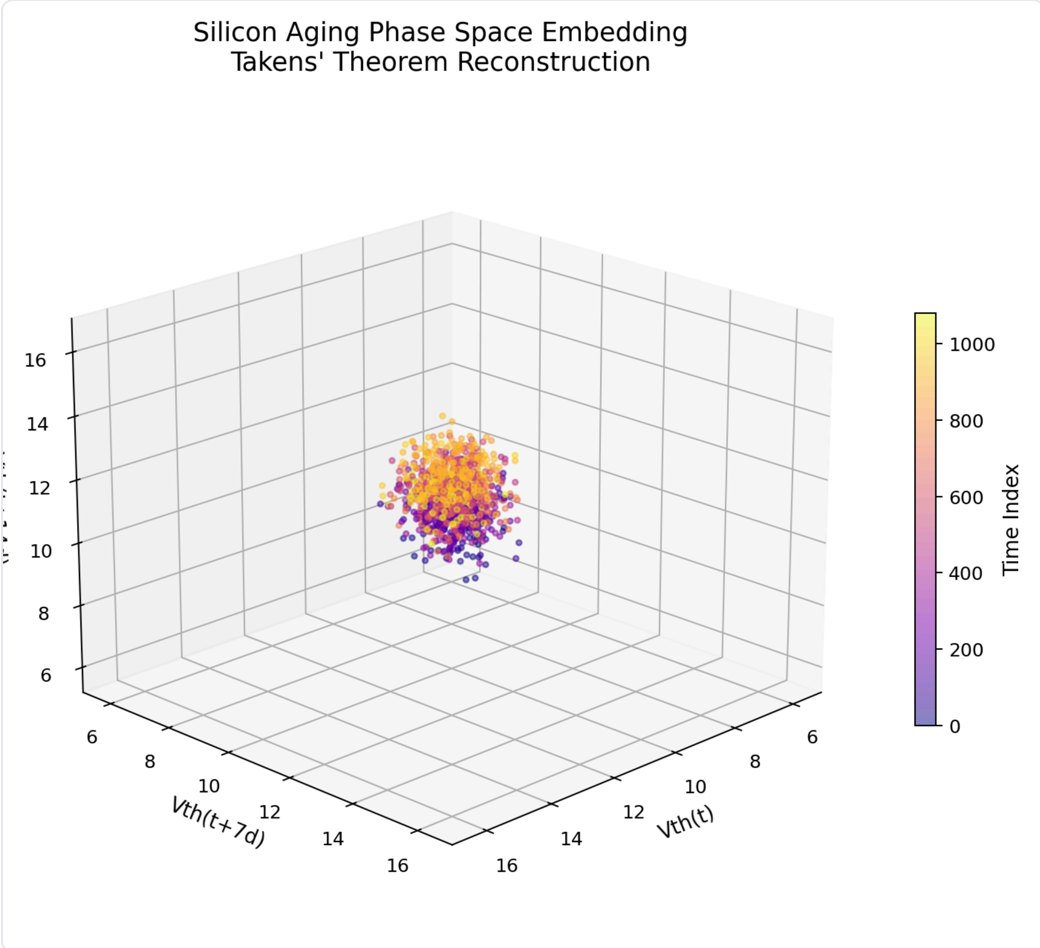
Project 1 — Datacenter Power Demand Visualizations



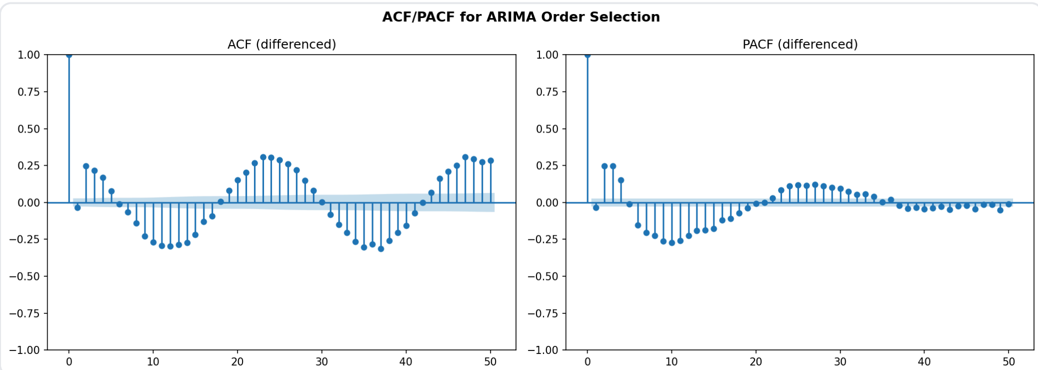
Proj1 Datacenter 3D Acf Landscape



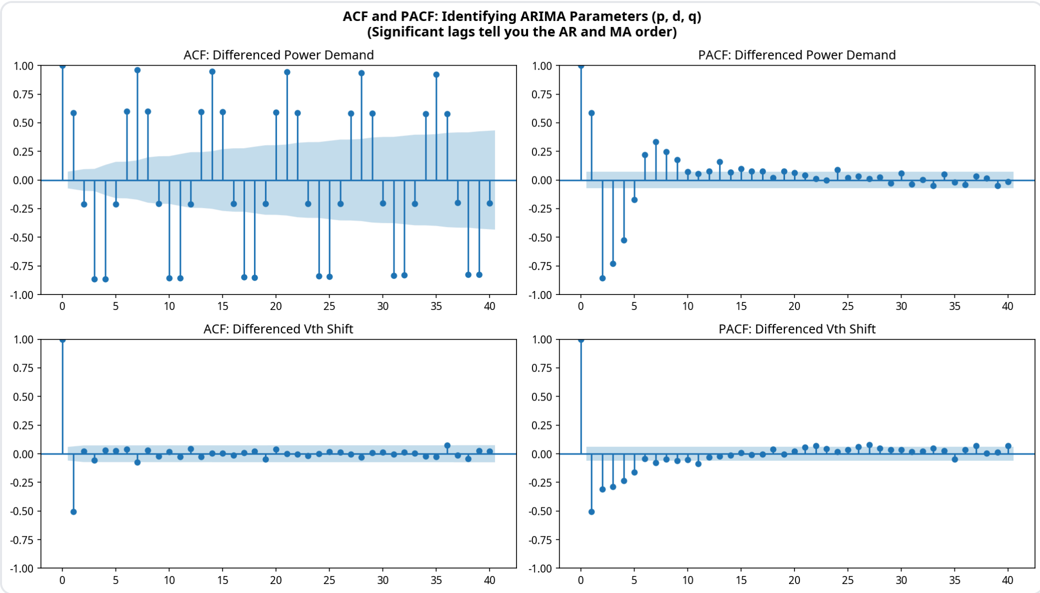
Proj1 Datacenter 3D Decomposition



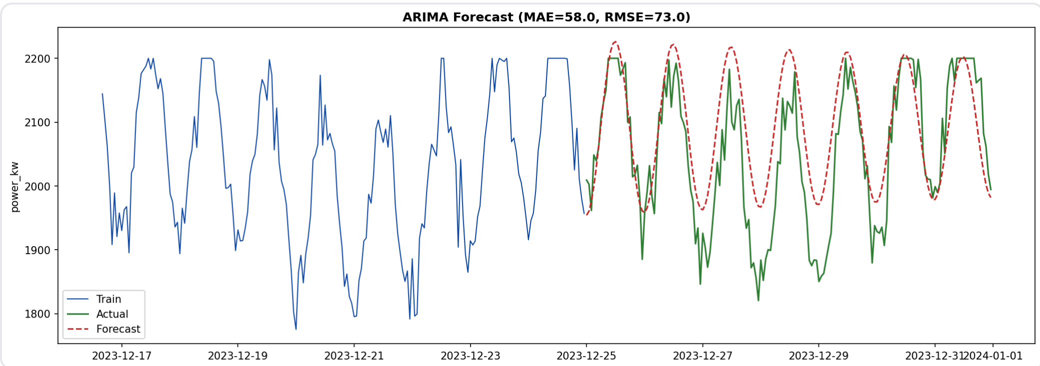
Proj1 Datacenter 3D Phase Space



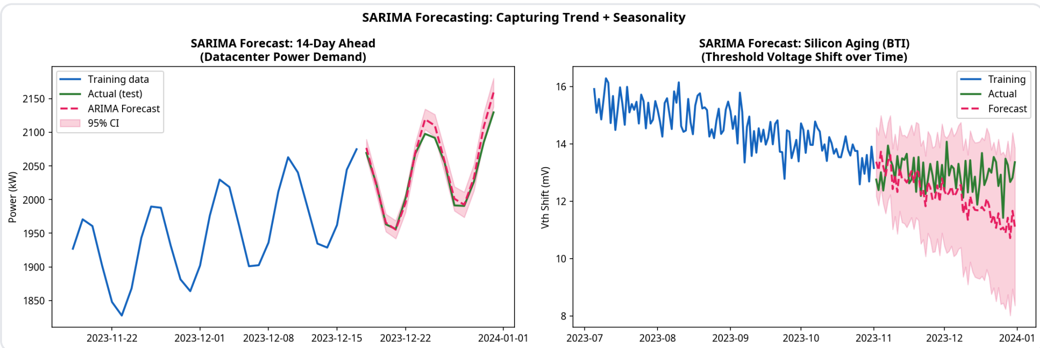
Proj1 Datacenter Acf Pacf



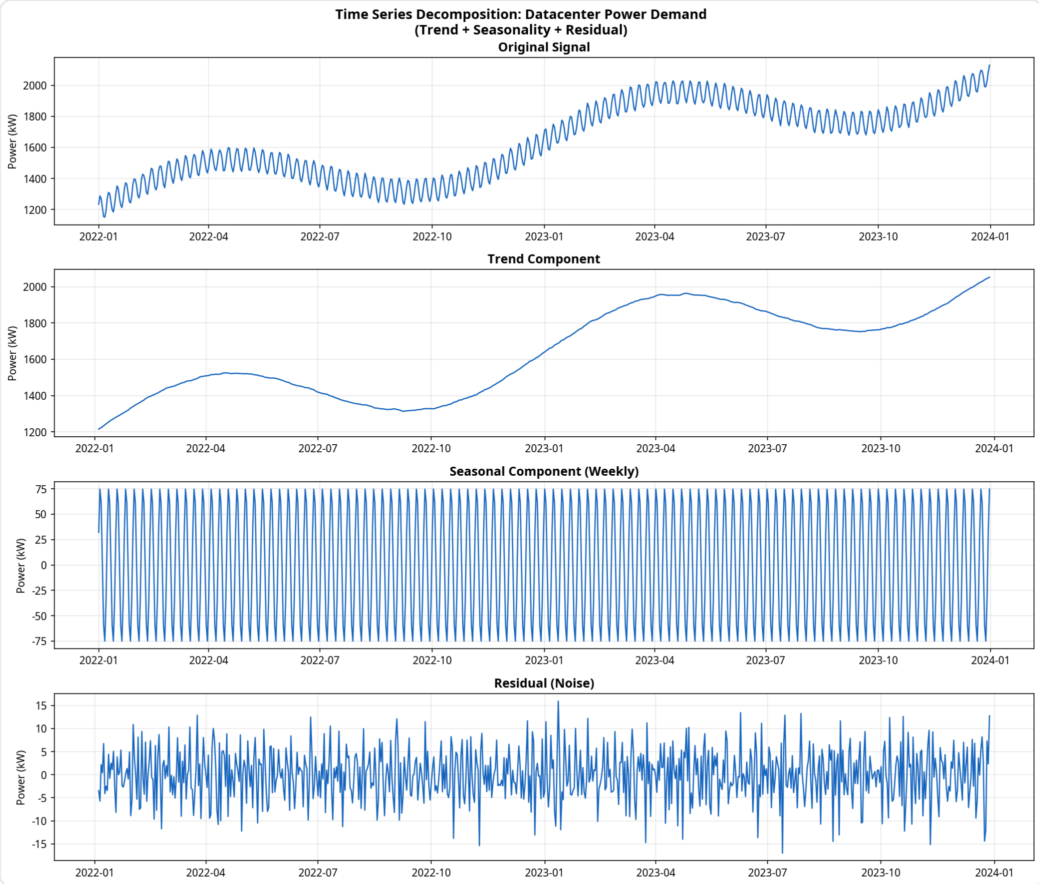
Proj1 Datacenter Acf Pacf Analysis



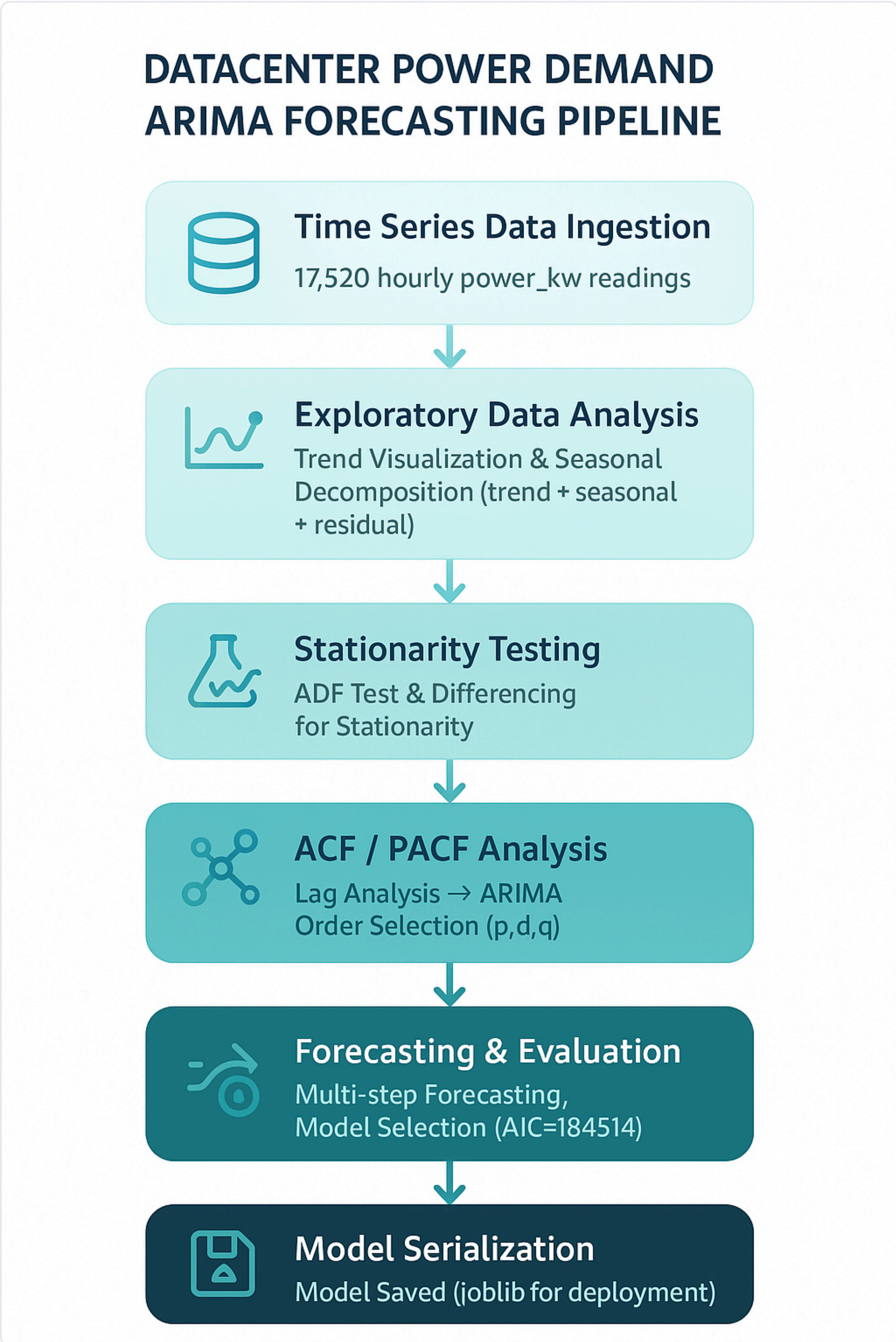
Proj1 Datacenter Arima Forecast



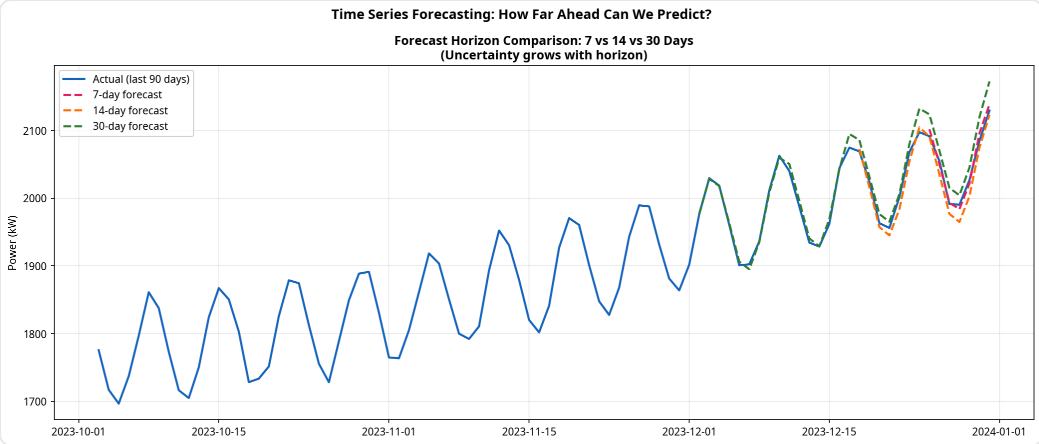
Proj1 Datacenter Arima Forecast Comparison



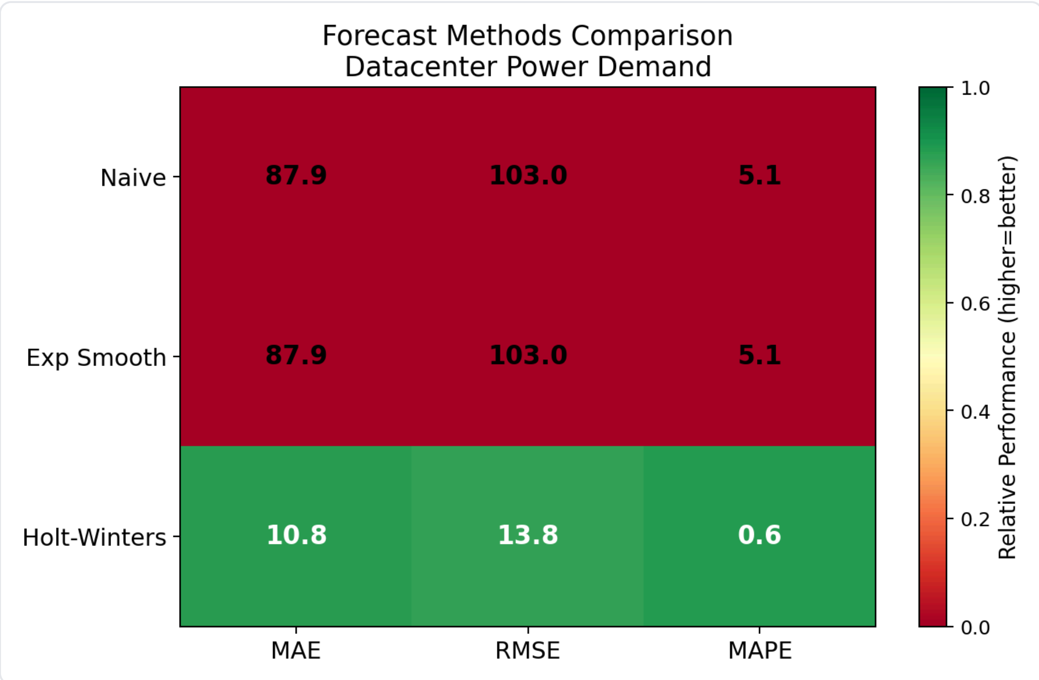
Proj1 Datacenter Decomposition



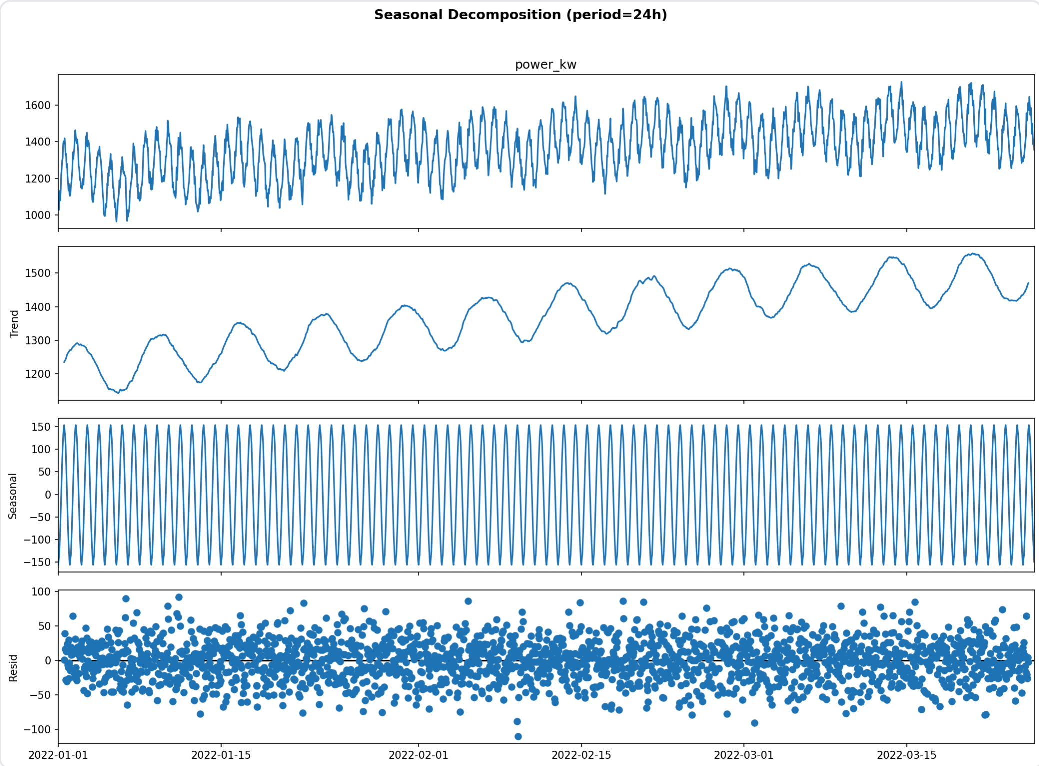
Proj1 Datacenter Flowchart



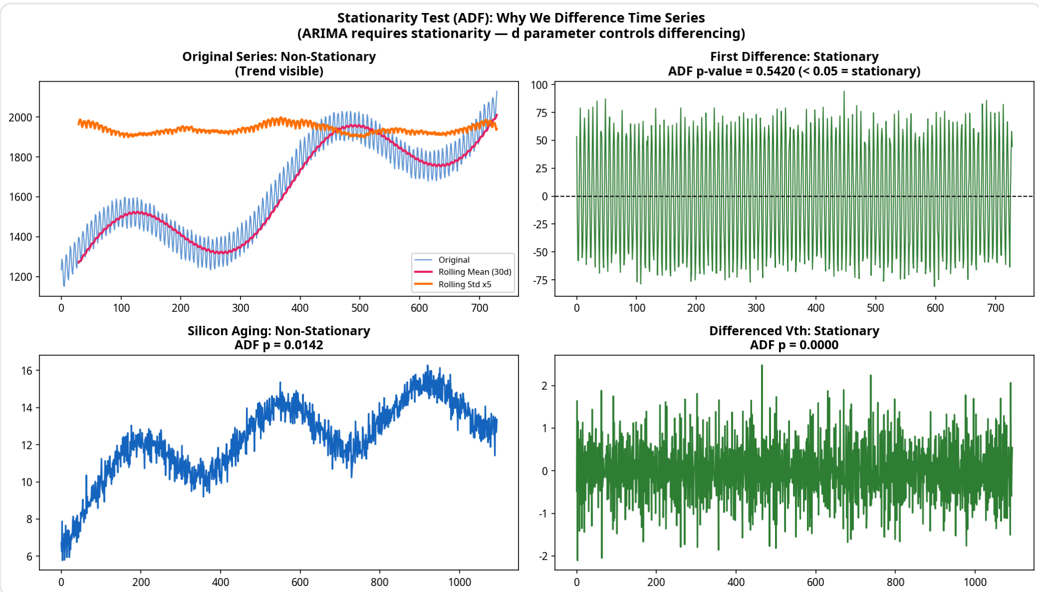
Proj1 Datacenter Forecast Horizons



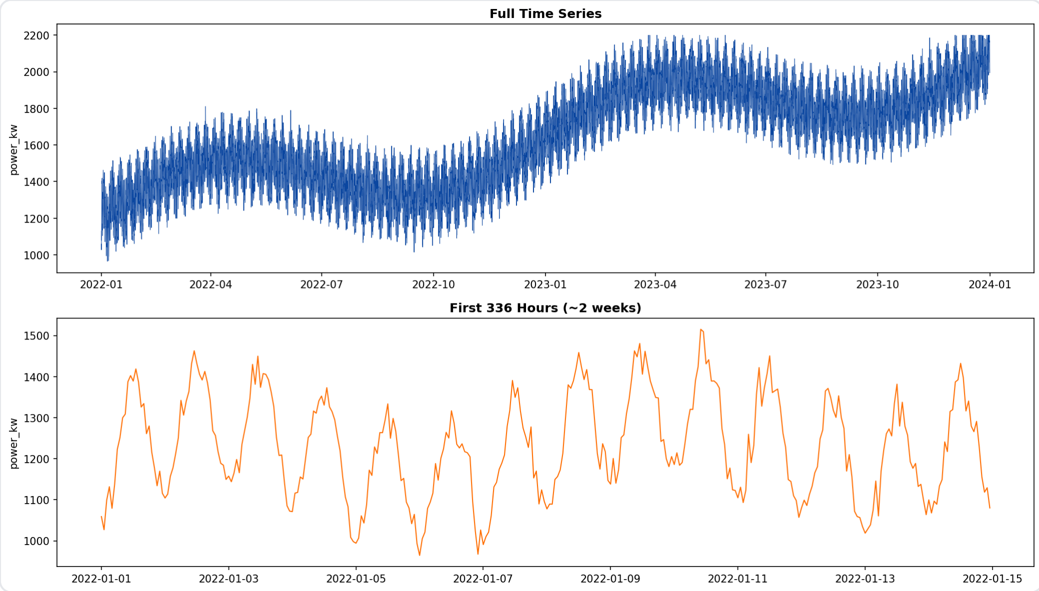
Proj1 Datacenter Model Heatmap



Proj1 Datacenter Seasonal Decomposition

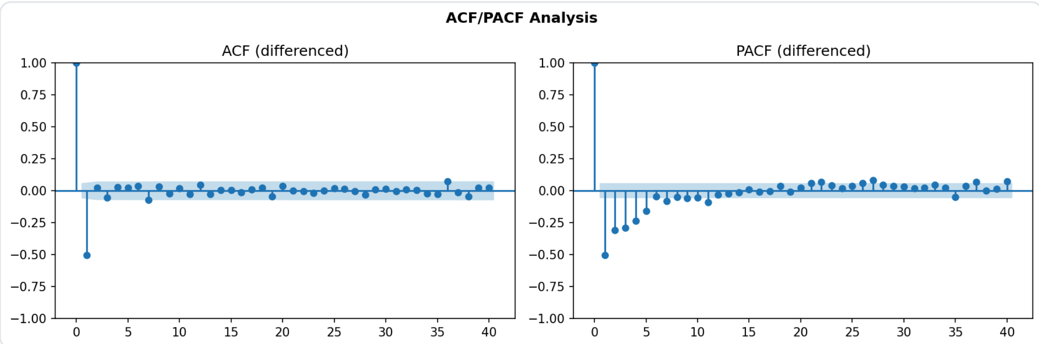


Proj1 Datacenter Stationarity

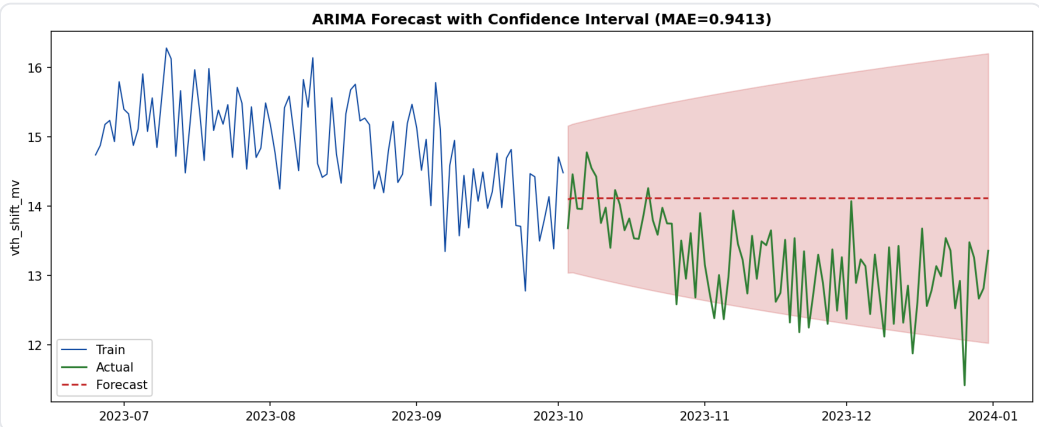


Proj1 Datacenter Ts Overview

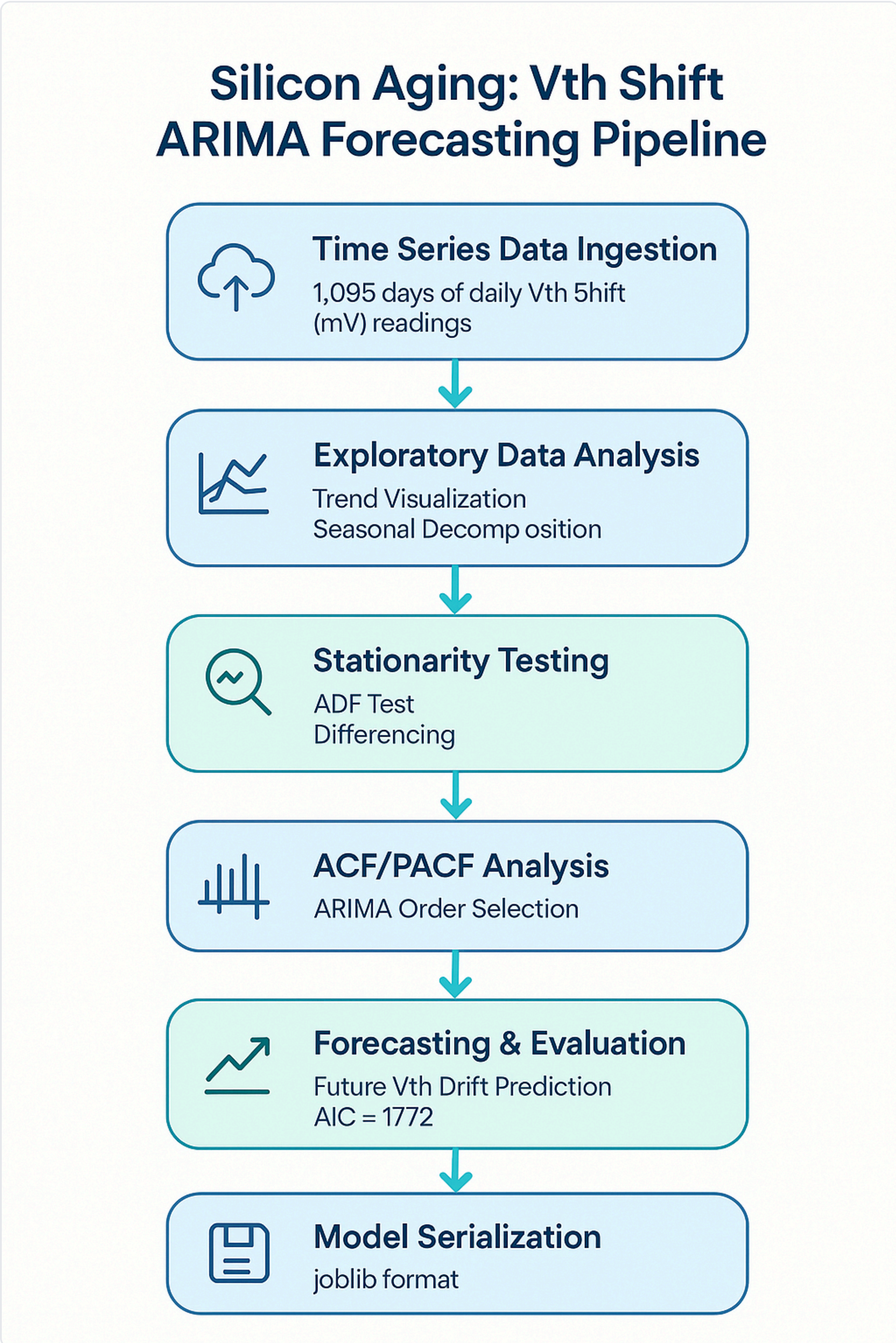
Project 2 — Silicon Aging Vth Shift Visualizations



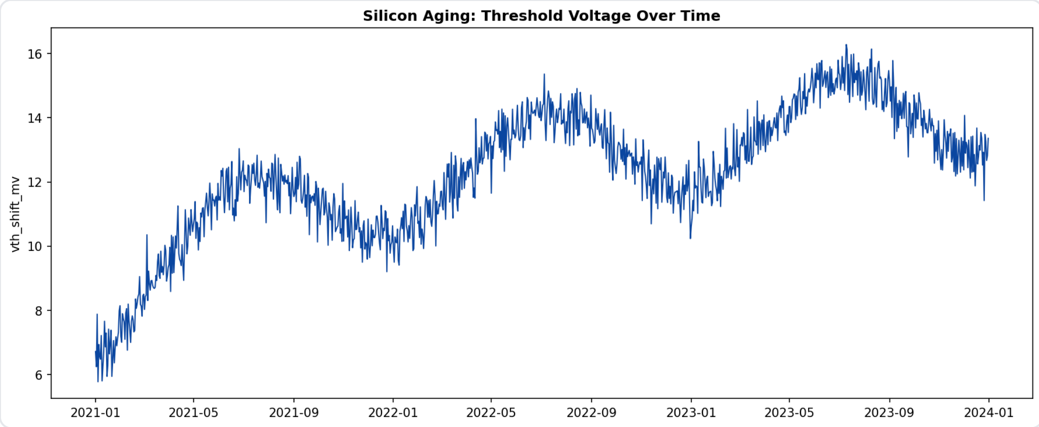
Proj2 Silicon Aging Acf Pacf



Proj2 Silicon Aging Arima Forecast



Proj2 Silicon Aging Flowchart



Proj2 Silicon Aging Ts Overview

Architecture

- **Training:** `src/train.py` — DATASETS dict pattern, trains both ARIMA models
- **Inference:** `src/predict.py` — multi-step forecast from any trained model
- **API:** `src/api.py` — FastAPI endpoint, POST steps → forecast values
- **Testing:** `tests/test_model.py` — 4 tests (existence + forecast × 2)